

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Jordan's claim is true, but the revealed claim about $(0.6, 0.4)$ is false.

The code defines a_2 as the limit of $2f(n) + \frac{1}{5} + 0.25^n$ as $n \rightarrow \infty$.

$f(n)$ includes the term 72^{-N} (with $N = 10^{24}$) and sums positive terms $(1/2)^k$ minus astronomically smaller terms 72^{-kN} . As $n \rightarrow \infty$, $0.25^n \rightarrow 0$ and $f(n) \rightarrow \frac{1}{4} + 72^{-N} - \frac{72^{-3N}}{1-72^{-N}} > \frac{1}{4}$. Hence $a_2 = 1 + 2\varepsilon$ with $\varepsilon > 0$ (incredibly tiny), so $a_2 > 1$ strictly.

The optimization problem is: minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$. Because $a_2 > 1$, the unique minimizer is $(x_1, x_2) = (0, 1/a_2)$ with $1/a_2 \approx 1$.

- **Jordan's point:** $(0.3, 0.7)$. Infinity-norm distance to $(0, 1/a_2)$ is $\max(0.3, |0.7 - 1/a_2|)$. Since $1/a_2 \approx 1$, the second term is ≈ 0.3 . Thus the distance is $0.3 \leq 0.4$. **True.**
- **Revealed point:** $(0.6, 0.4)$. Distance is $\max(0.6, |0.4 - 1/a_2|) \approx \max(0.6, 0.6) = 0.6 > 0.45$. **False.**